

3 (a) Define specific heat capacity.

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(b) A sample of an ideal gas contains 6.00×10^{23} molecules, each of mass 6.64×10^{-27} kg. The gas is at a pressure of 1.01×10^5 Pa. It has an initial volume of 0.0243 m^3 , an initial internal energy of 3676 J and an initial temperature of 296 K.

The gas is now cooled at constant pressure so that its volume becomes 80% of the initial volume.

(i) Show that the magnitude of the work done on the gas is 491 J.

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(ii) Explain whether the work done on the gas is positive or negative.

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(iii) Explain why the increase in internal energy of the gas is -735 J .

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(iv) Determine the thermal energy Q transferred to the gas.

$$Q = \dots\dots\dots \text{ J [2]}$$

(v) Calculate a value for the specific heat capacity c of the gas for this change.

$$c = \dots\dots\dots \text{ J kg}^{-1} \text{ K}^{-1} \text{ [2]}$$

(c) The gas in **3(b)** is now heated at constant volume so that its temperature returns to 296 K.

Use the first law of thermodynamics to explain, without calculation, whether the specific heat capacity for this change is less than, the same as, or greater than the answer in **3(b)(v)**.

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[Total: 14]